Product Sales - C++ Program

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

class Product {
private:
    int productId;
    string productName;
    double unitPrice;
    int monthlySales[12]; // Sales quantity for 12 months
    int totalQuantitySold;
    double totalBill;

public:
    void acceptProductData() {
        cout << "\nEnter Product ID: ";
        cin >> productId;
        cin.ignore();
        cout << "Enter Product Name: ";
        getline(cin, productName);
        cout << "Enter Unit Price ($): ";
        cin >> unitPrice;
        totalQuantitySold = 0;
        cout << "Enter sales quantity for 12 months:\n";
        for (int i = 0; i < 12; i++) {
            cout << " Month " << (i + 1) << ": ";
            cin >> monthlySales[i];
            totalQuantitySold += monthlySales[i];
        }
        totalBill = totalQuantitySold * unitPrice;
    }

    int getTotalQuantity() const { return totalQuantitySold; }
    double getTotalBill() const { return totalBill; }

    void displayProduct() const {
        cout << left << setw(10) << productId
             << setw(20) << productName
             << setw(12) << fixed << setprecision(2) << unitPrice
             << setw(15) << totalQuantitySold
             << setw(15) << totalBill << "\n";
    }
};

int main() {
    int n;
    cout << "Enter total number of products: ";
    cin >> n;
    Product products[50]; // Array of Product objects
    for (int i = 0; i < n; i++) {
        cout << "\n--- Input Details for Product " << (i + 1) << " ---";
        products[i].acceptProductData();
    }

    double grandTotalBill = 0;
    int grandTotalQuantity = 0;
    cout << "\n================================ PRODUCT SALES SUMMARY ================================\n";
    cout << left << setw(10) << "ID"
```

Made with Julius

Product Sales - C++ Program

```
         << setw(20) << "Name"
         << setw(12) << "Price ($)"
         << setw(15) << "Qty Sold"
         << setw(15) << "Total Bill ($)" << "\n";
    cout << "-------------------------------------------------------------------------------------\n";
    for (int i = 0; i < n; i++) {
        products[i].displayProduct();
        grandTotalQuantity += products[i].getTotalQuantity();
        grandTotalBill += products[i].getTotalBill();
    }
    cout << "-------------------------------------------------------------------------------------\n";
    cout << "Grand Total Quantity Sold : " << grandTotalQuantity << " units\n";
    cout << "Grand Total Bill / Revenue: $" << fixed << setprecision(2) << grandTotalBill << "\n";
    cout << "=====================================================================================\n";
    return 0;
}
```

Made with Julius